BURST MONITOR

Dashboard User Manual

Phase-Coherence Topology × S&P 500

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Resolution Geometry

March 2026

BURST MONITOR v2.0

Phase-Coherence Topology × S&P 500 × Signature Library — Data through 2026-03-23

NEUTRAL | CORRECTION-PRONE

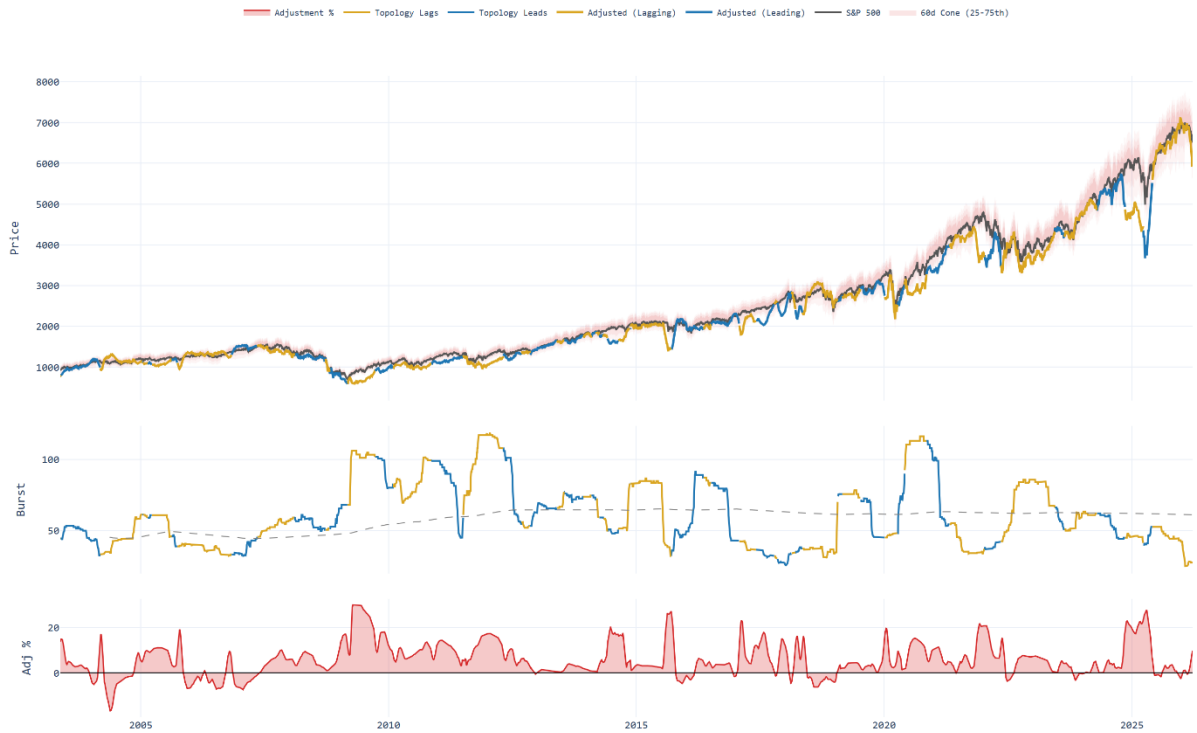
P(neg) badge: historical analog match
Structural State: current phase geometry
These signals measure different dimensions and may diverge.

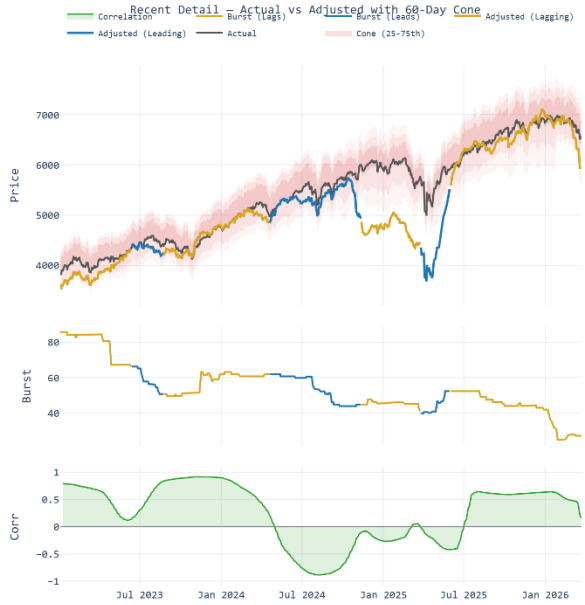
S&P 500	ADJUSTED	GAP	BURST Z	REGIME	CORR	DECOUPLE	TRIGGER
6,581	5,939	9.8%	-1.48	LAGGING → LEADING	-0.688	0.46	0.67
		642 pts	raw = 27	(regime transition in progress)			CORRECTION-PRONE

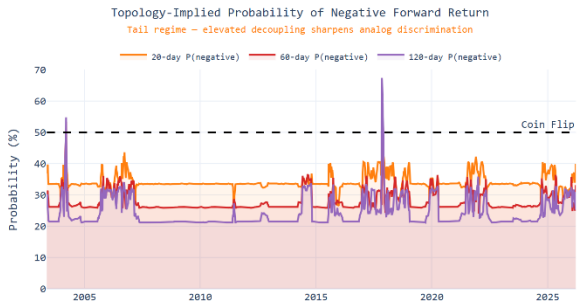
SIGNATURES: S1 Floor Loss | S2 Corr Flip | S3 Coupled Cascade | S4 False Alarm | S5 Recovery

Current State

The burst-adjusted S&P 500 with topology-implied confidence cone. Blue = topology leading price. Gold = topology lagging. Cone bands show the historical outcome distribution at similar decoupling levels — scenario envelope, not a forecast.

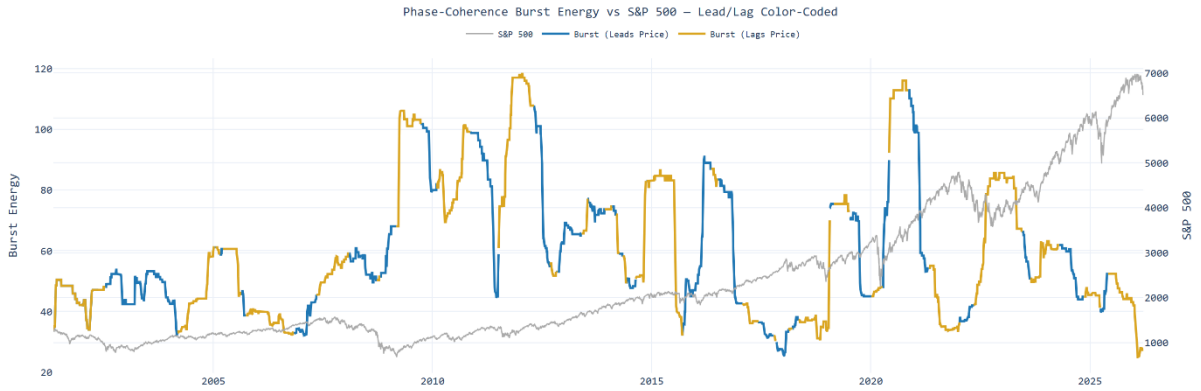


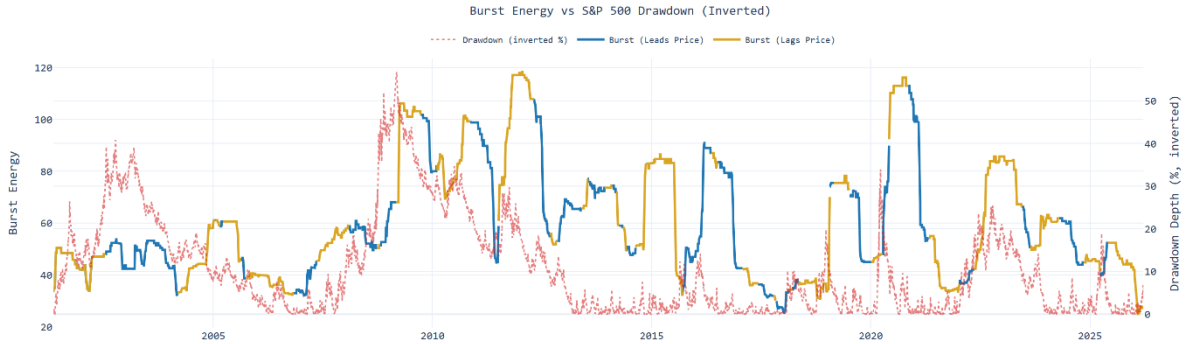




The Raw Topology

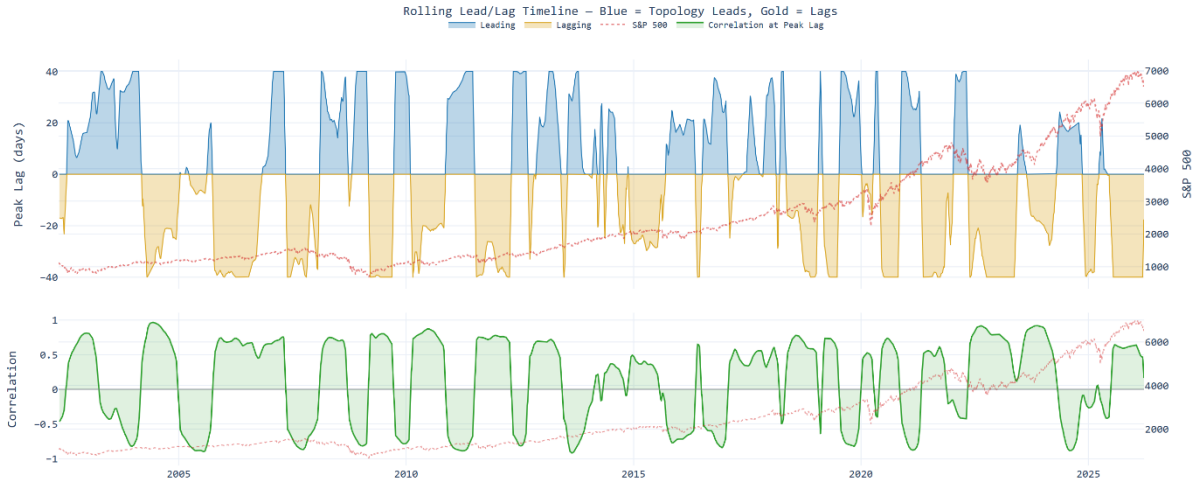
Phase-Coherence Burst Energy measures how many independent degrees of freedom the market actually has.

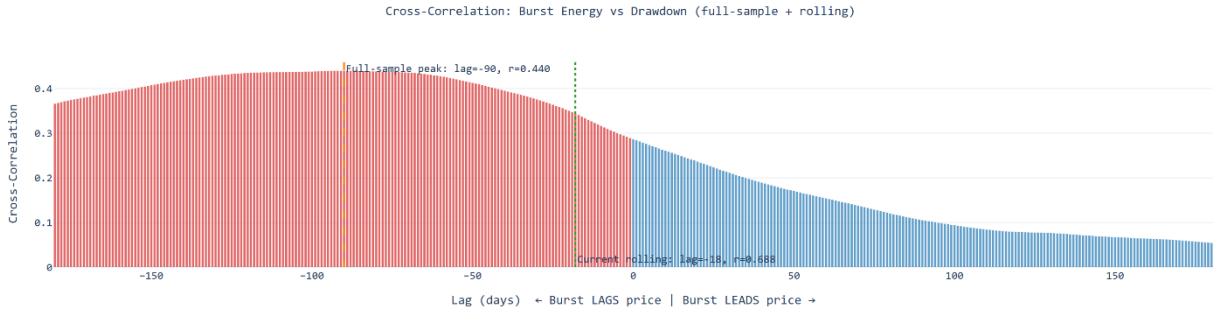




The Lead/Lag Structure

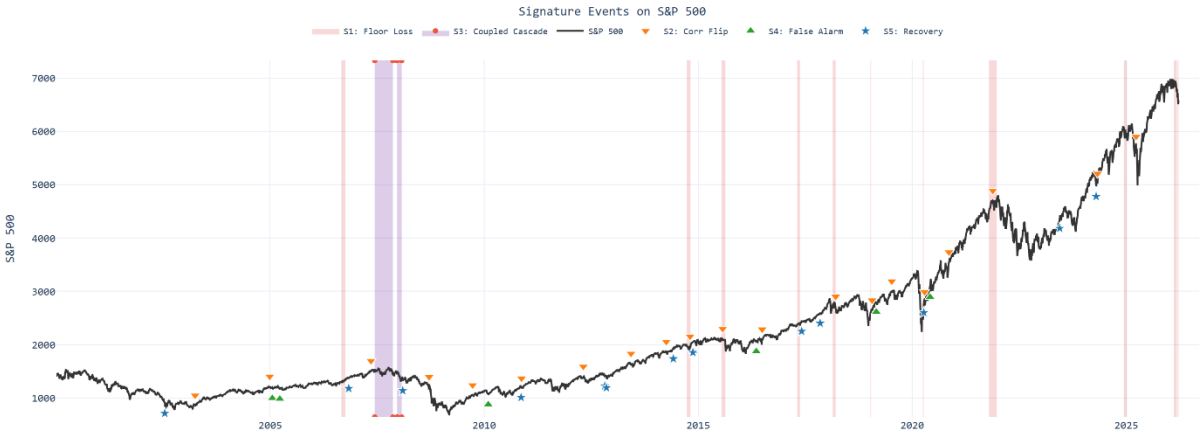
Rolling cross-correlation reveals whether topology is warning ahead of price (leading) or registering damage after the fact (lagging).

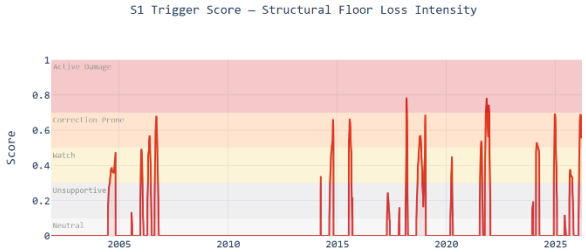


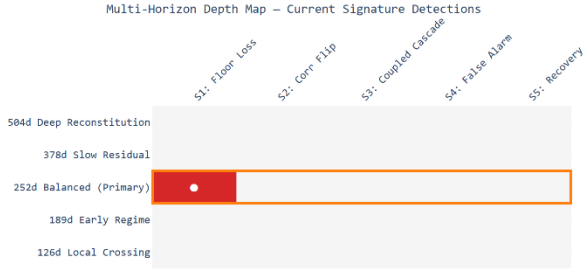


Signature Library

Five topological signatures detect distinct structural transitions. S1 (Floor Loss) and S3 (Coupled Cascade) are danger signals. S4 (False Alarm) and S5 (Recovery) are safe/all-clear signals. S2 (Corr Flip) is the broad base layer.

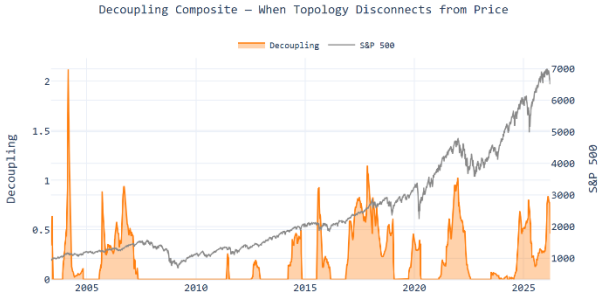


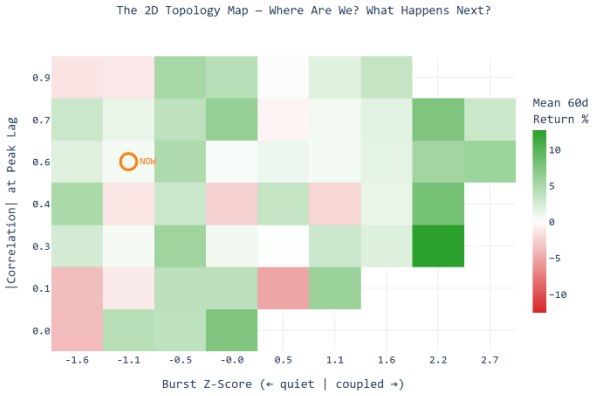




The Decoupling Gradient

When burst energy drops AND correlation weakens simultaneously, the topology has disconnected from price. Note: When Burst Z > 0 (active structural coupling), Decoupling Composite = 0 by construction and P(neg) defaults to the unconditional base rate — P(neg) is a pre-crisis signal, not a during-crisis severity indicator.





What This Dashboard Shows

The Burst Monitor is a system that looks at the S&P 500 from a fundamentally different angle than traditional financial analysis. Instead of studying prices, earnings, or sentiment, it studies the hidden geometric structure underneath the market — the topology of how individual stocks move in relation to each other.

Think of it this way: every trading day, hundreds of stocks are rising, falling, and oscillating. Each stock has its own rhythm, its own cycle. Sometimes those cycles line up — stocks close their cycles together, like a crowd synchronizing their footsteps on a bridge. The Burst Monitor detects and measures that synchronization.

When many stocks complete their cycles at the same time, that is a closure event. The energy of those closure events — how strong and concentrated the synchronization is — is what we call Phase-Coherence Burst Energy. It turns out that this measurement, which has nothing to do with price directly, contains information about where the market is headed.

The dashboard presents this information in layers, from the headline summary at the top to the deep structural detail at the bottom. This manual walks through every element, top to bottom, explaining what it shows, how it is computed, and what it means for you as a reader.

Core Concepts Before You Begin

Phase-Coherence Burst Energy

Every stock has a price cycle. Using a mathematical technique called the Hilbert transform, we extract the phase (position within its cycle) of each stock on each day. Phase is measured from 0 to 1, where 0 (or equivalently, 1) means the stock is completing a cycle — it has "closed."

A closure event happens when an unusually large number of stocks close their cycles on the same day. To measure how unusual this is, we build a histogram of where all stocks sit in their cycles, then measure how far that histogram deviates from a uniform (random) distribution. The mathematical measure is the L2 norm of the deviation. A high value means the market is highly synchronized; a low value means stocks are moving independently.

The dashboard tracks this value over time and converts it to a z-score (number of standard deviations above or below the historical average). A z-score of zero means average synchronization. Positive means unusually coupled. Negative means unusually quiet or decoupled.

> **Why this matters**
>
> Market crashes and major corrections are preceded by structural changes in how stocks co-move. Burst energy captures this structural dimension before it shows up in price. A market where stocks are losing their coordination is structurally weaker than one where they are synchronized, even if prices look the same.

Lead/Lag

The system continuously measures whether changes in burst energy tend to happen before or after changes in the S&P 500. It does this by computing a rolling cross-correlation — sliding one signal forward and backward in time and measuring the alignment at each offset.

If burst energy changes best predict the S&P 500 when shifted forward by, say, 60 days, then topology is "leading" price by 60 days. If the best alignment requires shifting burst energy backward, then topology is "lagging" — it is registering damage after the fact rather than warning ahead.

On the dashboard, **blue** always means the topology is leading (warning ahead of price), and **gold** means lagging (registering damage after). This color convention is used throughout every chart.

Correlation

The correlation value tells you how strong the relationship between burst energy and the S&P 500 is at the current lead/lag offset. A correlation of 0.8 means the signals are tightly coupled; the burst topology is highly informative about price. A correlation of 0.3 means the relationship is weak; the topology exists but has less to say about near-term price action.

The sign of the correlation matters too. A positive correlation means higher burst energy corresponds to larger drawdowns (the topology is measuring stress). A negative correlation means the inverse relationship holds.

Decoupling

Decoupling is the danger signal. It is what happens when burst energy drops (the topology goes quiet) AND the correlation between topology and price weakens simultaneously. This means the structural scaffolding that normally connects the internal dynamics of the market to its price has broken down.

The Decoupling Composite is calculated as: $\max(0, -\text{burst_z}) \times (1 - |\text{correlation}|)$. It is zero when things are normal (either burst z is positive or correlation is strong). It rises when burst energy is below average AND the linkage to price is weak. Historically, high decoupling readings precede the worst forward returns in the dataset.

The Confidence Cone

The confidence cone is not a prediction. It is a histogram of what happened historically when the market was in a similar topological state. The system looks at the current decoupling composite value, finds all historical days where the decoupling composite was within a similar range, and then looks at what the S&P 500 actually did over the following 20, 60, and 120 trading days.

The cone shows percentile bands — for example, the 10th to 90th percentile of historical outcomes. A narrow cone means historical analogs are tightly clustered (the topology has a strong opinion about what happens next). A wide cone means history is ambiguous from this state.

The Header and Status Badge

At the very top of the dashboard, you will see a title bar and a colored status badge in the upper right corner. The badge provides a single-word assessment of current conditions based on the probability of negative forward returns over the next 60 trading days.

In v2.0, the dashboard shows both the original P(neg) badge and a new Structural State badge derived from the signature engine. These two badges measure different things and can diverge. The P(neg) badge reflects historical analog matching — it asks, “when the market was in a similar topological state before, how often did it decline?” The Structural State badge reflects the current phase geometry configuration — it asks, “are the topological signatures indicating structural deterioration right now?”

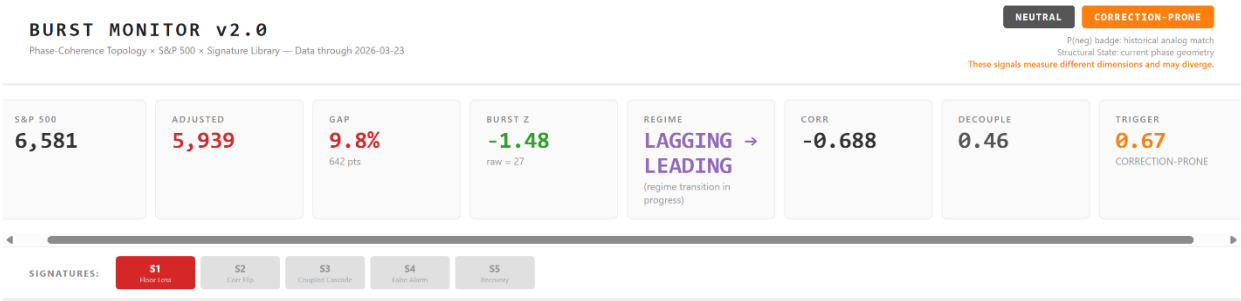
A subtitle beneath the badges explains this distinction: “P(neg) badge: historical analog match — Structural State: current phase geometry configuration.” When these two assessments disagree (for example, P(neg) reads NEUTRAL while the Structural State reads CORRECTION-PRONE), the dashboard displays a note: “These signals measure different dimensions of market structure and may diverge.” This is not a contradiction — it means the analog history and the live geometry are telling different stories, which is itself informative. The Structural State badge is generally more responsive to current conditions because it reads the geometry directly, while the P(neg) badge is constrained by the quality of its analog pool (see the confidence qualifier discussion under the P(60d NEG) stat card).

Below the structural state badge, five small indicator lights labeled S1 through S5 show which individual signatures are currently active at the baseline (252-day) horizon. Each light is gray when inactive and colored when active. Hovering over each indicator shows a brief tooltip describing what that signature detects.

Status Badge Levels P (neg)

Badge	Trigger	Meaning
ELEVATED RISK	P(neg) > 45%	More than 45% of historical analogs from the current topological state resulted in negative 60-day forward returns. The topology is in a configuration that has historically preceded poor outcomes.
CAUTION	P(neg) > 35%	Between 35–45% of analogs were negative. The topology is showing strain but has not reached the worst historical configurations.
SUPPORTED	Adj < −2%	The burst-adjusted price is more than 2% above the actual S&P 500, meaning the topology suggests the market is undervalued relative to its structural state.
NEUTRAL	Default	None of the above conditions are met. The topology is not sending a strong signal in either direction.

The Stat Cards

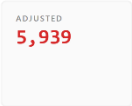


Directly below the header is a row of eight stat cards. These are the key numbers, updated with each pipeline run. Here is what each one reports:

S&P 500

The current closing price of the S&P 500 index. This is the actual, unmodified market price. It is your anchor — everything else on the dashboard is measured relative to this number.

ADJUSTED



The burst-adjusted S&P 500 price. This is what the market "should" be priced at according to the topology. It is computed by applying a daily adjustment percentage (see GAP below) to the actual S&P 500.

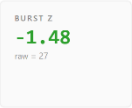
The adjustment integrates four signals: the burst energy z-score (how active the topology is), a regime modifier (amplified when the lead/lag is large), the correlation confidence (how much weight to give the topology signal), and a decoupling penalty (applied when the topology disconnects from price). These four channels are combined, scaled, clipped to ±30%, and smoothed over 21 trading days.

When the adjusted price is above the actual price, the topology says the market is underpriced. When it is below, the topology says the market is overpriced.

GAP

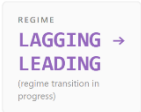
The percentage difference between the adjusted and actual S&P 500, along with the point difference. A gap of −2.2% with −144 points means the adjusted price is 2.2% (144 points) above the actual price, suggesting topological support. The sign convention: negative means the adjusted is above actual (support), positive means adjusted is below actual (stress). The card turns red when the gap exceeds +2%, indicating the topology is flagging overvaluation.

BURST



The current Phase-Coherence Burst Energy value and its z-score. The raw number (e.g., 27.0) is the L2 norm of the phase histogram deviation on the most recent closure event. The z-score (e.g., z = −1.50) tells you how that compares to the trailing one-year average. A z-score of zero is average. Below zero means the market is quieter than normal — fewer stocks are synchronizing. Above zero means the topology is more active than usual.

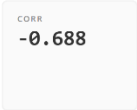
REGIME



Shows whether the topology is currently **LEADING** (anticipating price moves) or **LAGGING** (reacting to them). The number below (e.g., −73d) is the smoothed lead/lag in trading days. A value of +60 means burst energy changes precede S&P 500 changes by about 60 days. A value of −80 means burst energy is reflecting damage that happened about 80 days ago. The smoothed value uses a 21-day rolling average to prevent misleading single-day jumps.

When the topology is in the process of flipping regimes, the stat card shows a transition indicator. This happens when the raw (unsmoothed) lead/lag has already changed sign but the 21-day smoothed value has not caught up yet. During this window, the card displays an arrow notation — for example, “LAGGING → LEADING” — with the label “(regime transition in progress)” beneath it. The card’s background color blends between gold and blue during this period. This transition state typically lasts one to three weeks as the smoother catches up to the raw signal. It is worth noting because a regime flip is structurally meaningful: it marks the moment when the topology’s temporal relationship with price is reversing.

CORR

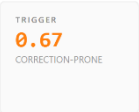


The correlation between burst energy and S&P 500 drawdown at the current peak lag offset. Values near ±1.0 mean the topology and price are tightly coupled. Values near 0 mean the relationship is weak. This number feeds directly into the confidence weighting of the adjustment and the decoupling composite. When correlation is strong, the system has more confidence in its adjustment.

DECOUPLE

The Decoupling Composite value. This is the core danger metric. It is computed as max(0, −burst_z) × (1 − |correlation|). It is zero when things are normal. It rises when burst energy is below average AND the correlation to price is weak — meaning the topology has disconnected. Values above 0.5 are concerning. The card turns red above 0.5. Historically, elevated decoupling composite readings have preceded the worst short-term forward returns.

TRIGGER



The five levels are: Neutral (no signatures of concern active), Structurally Unsupportive (weak pattern present but not operationally significant), Trigger Watch (pattern present and intensifying), Correction-Prone (strong Floor Loss pattern, historical analogs show elevated correction risk), and Active Damage (the most intense reading, typically seen during or immediately before significant drawdowns).

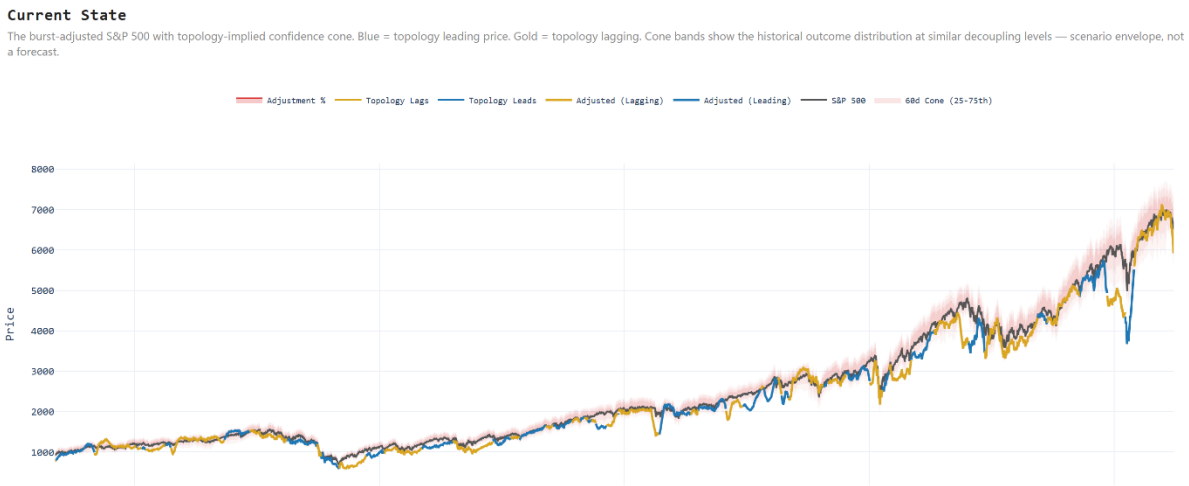
Below the badge, five small indicator lights labeled S1 through S5 show which individual signatures are currently active at the baseline (252-day) horizon. A lit indicator means that signature's conditions are currently met. The TRIGGER stat card replaces the P(60d NEG) card from v1.0 and shows the current S1 trigger score along with its taxonomy band.

Section 1: Current State

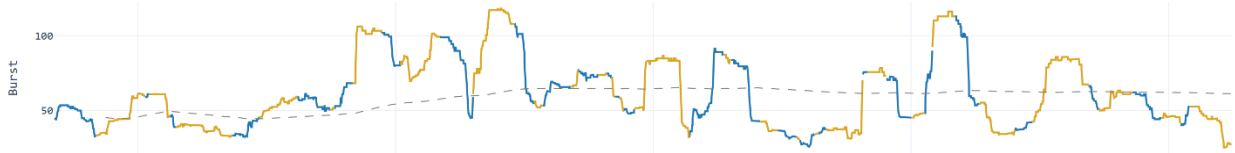
The first section of the dashboard contains three charts: the full-history adjusted view, a recent detail zoom, and the probability chart. Together they answer the question: where is the market relative to what the topology says, and how concerned should we be?

The Headline Chart: S&P 500 Actual vs Burst-Adjusted

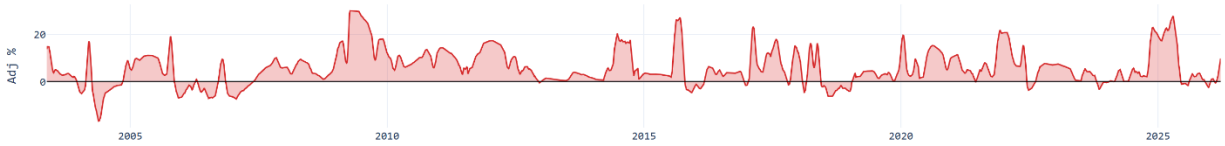
This is the main chart. It has three panels stacked vertically.



Top panel: The gray line is the actual S&P 500. The red line is the burst-adjusted S&P 500. When the red line diverges above the gray line, the topology suggests the market is undervalued relative to its structural state. When the red line drops below, the topology suggests overvaluation. The red-shaded bands around the lines are the confidence cone — they show the range of historical outcomes at the current topological state. A wider cone means more uncertainty; a narrower cone means the topology has a stronger directional opinion.

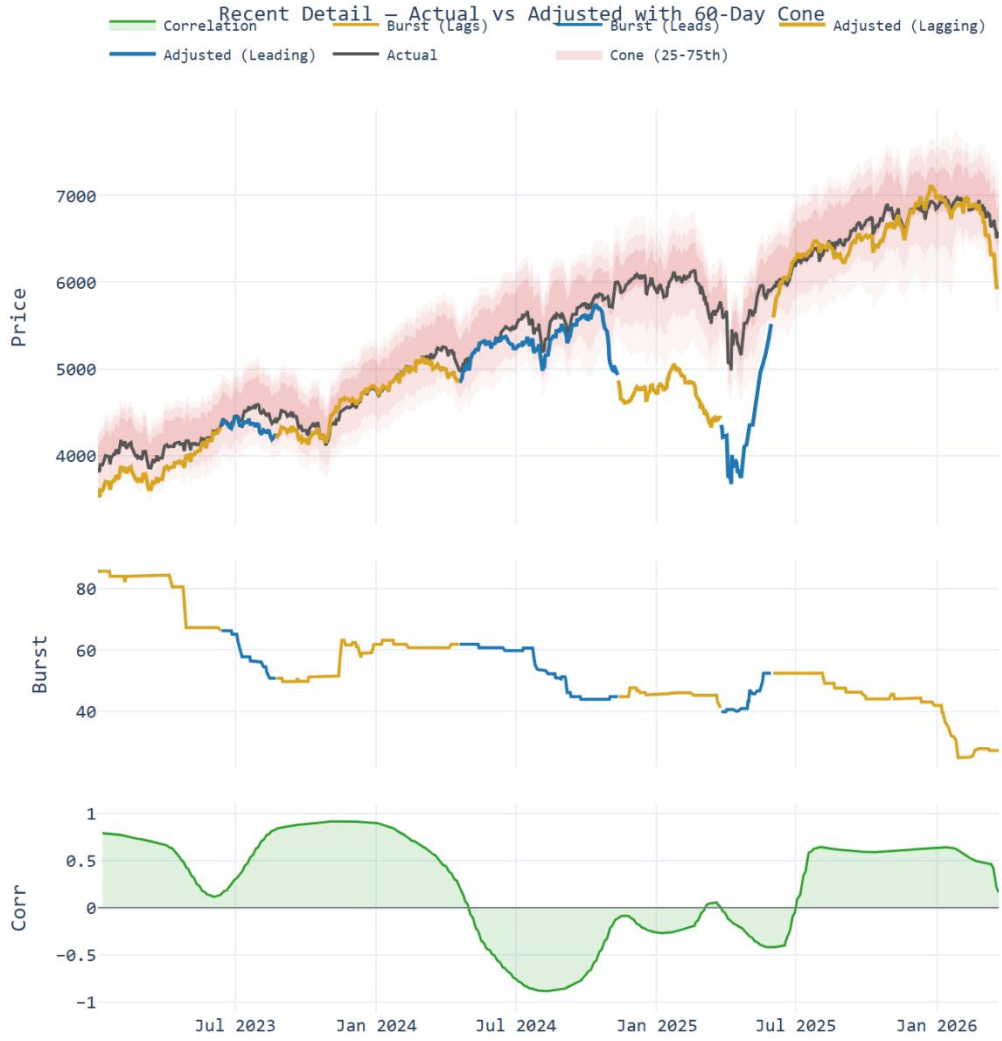


Middle panel: Burst energy over time, color-coded by lead/lag regime. **Blue** segments are periods where the topology was leading price (predictive). **Gold** segments are where it was lagging (reactive). The dashed gray line is the expanding historical mean. When burst energy is above the mean, the market is topologically active. Below the mean, it is quiet.



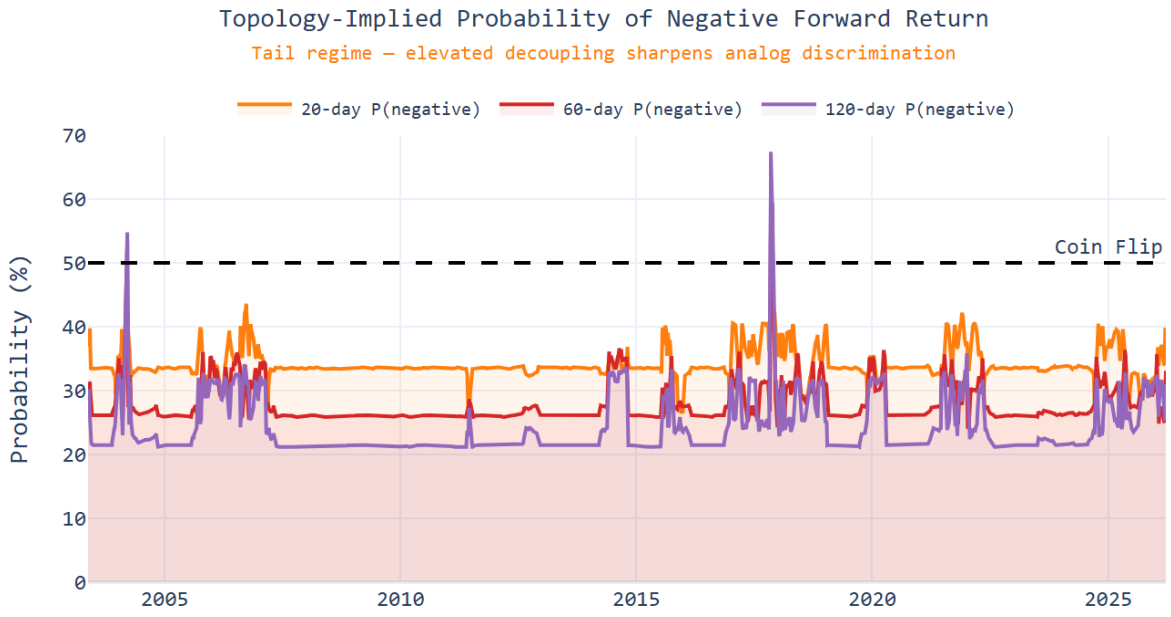
Bottom panel: The daily adjustment percentage applied to the S&P 500 to produce the adjusted line. Positive values push the adjusted price above the actual (topology sees stress). Negative values pull it below (topology sees support). This is the raw engine output before it accumulates into the price gap.

Recent Detail



This chart zooms into the last two to three years, showing the same actual vs. adjusted comparison with the confidence cone, but at a scale where recent movements are easier to read. The bottom sub-panel overlays burst energy and the smoothed correlation, giving you a quick sense of whether the topology is active and whether it is coupled to price.

Probability of Negative Forward Return



Important context: the confidence cone and the probability chart show a scenario envelope, not a forecast. They represent the distribution of historical outcomes when the market was in a similar topological state — a conditional histogram, not a point prediction. Two conditions reduce the informativeness of these readings. First, when the decoupling composite is below 0.3, the analog pool is not differentiated enough to produce meaningful probabilities; the chart subtitle will note this. Second, when Burst Z is above zero (active structural coupling), the decoupling composite is zero by construction, so the probability defaults to the unconditional base rate; a warning will note that P(neg) is uninformative during active structural events and to refer to the Structural State instead.

This chart shows the historical probability of negative S&P 500 returns at three horizons: 20 trading days (roughly one month), 60 trading days (roughly one quarter), and 120 trading days (roughly six months). Each line is computed by finding historical analogs — days where the decoupling composite was similar to today — and counting what fraction experienced a negative return at each horizon.

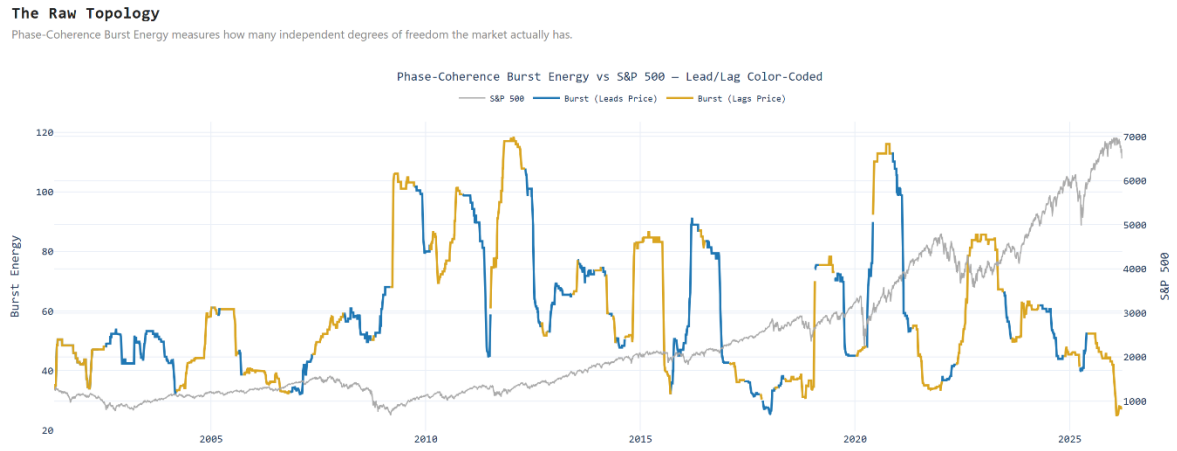
The dashed horizontal line at 50% is labeled "Coin Flip." When the probability curves are below this line, historical analogs suggest the market was more likely to go up than down from this topological state. When they rise above it, the historical odds have flipped unfavorable. The further above 50%, the more consistently poor the historical outcomes from similar conditions.

> **How to read the three horizons together**
>
> If the 20-day probability is high but the 120-day probability is low, it suggests a near-term drawdown risk that typically recovers within six months. If all three horizons are elevated, it suggests a more sustained period of poor returns. If the 120-day is higher than the 20-day, historical analogs suggest the pain comes later, not immediately — a slow-building problem rather than a sudden shock.

Section 2: The Raw Topology

This section steps behind the adjusted price and shows you the raw signal: burst energy itself. These charts help you understand what the topology is actually doing before it gets translated into an adjustment.

Burst Energy vs S&P 500

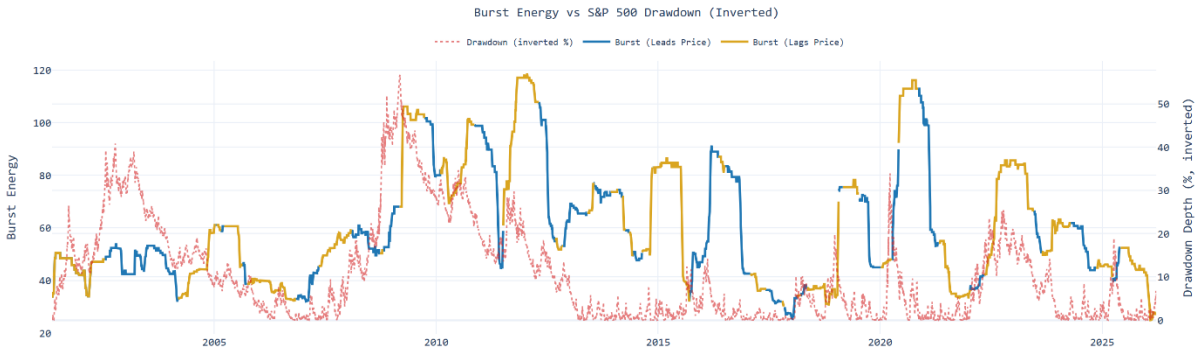


This dual-axis chart plots burst energy (left axis) against the S&P 500 (right axis, gray line). The burst energy line is color-coded by lead/lag regime: blue when the topology is leading price, gold when lagging.

What to look for: during major market events, you will see burst energy spike — these are periods of extreme phase synchronization, where the topology is highly active. In the 2008 financial crisis and the 2020 COVID crash, burst energy surged dramatically. The color tells you whether the topology was warning ahead of the move (blue) or registering the damage after (gold).

During calm markets, burst energy tends to be low and quiet. Extended periods of low burst energy are not necessarily good — they can indicate the topology is dormant, which means the correlation to price weakens and the system has less to say about near-term direction.

Burst Energy vs Drawdown



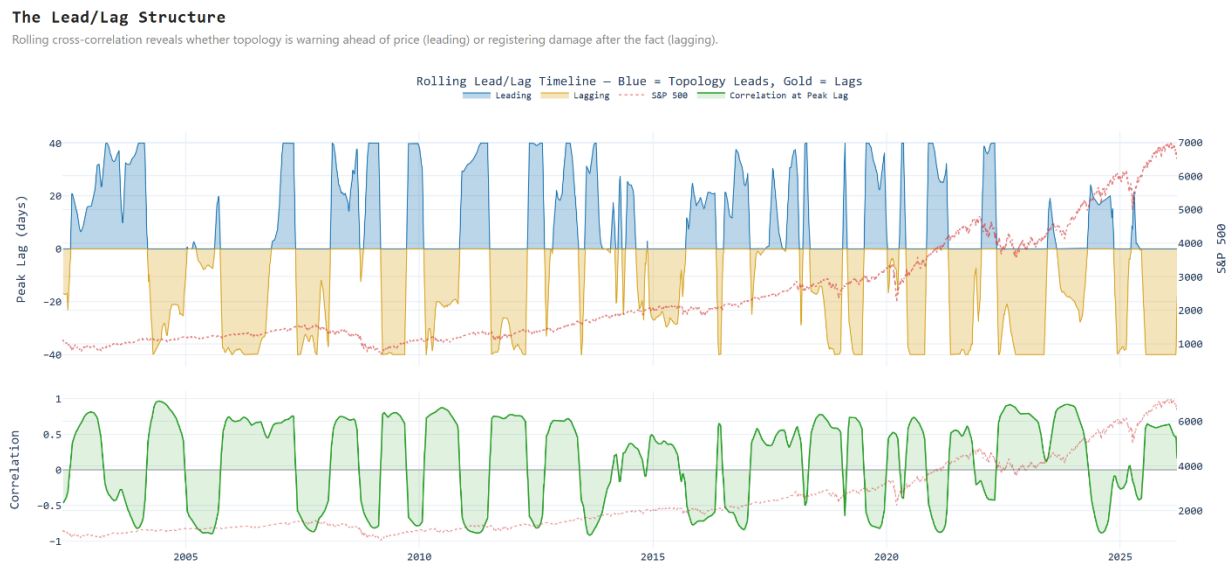
This chart shows the same burst energy signal but plotted against the S&P 500 drawdown (inverted so that deeper drawdowns appear as higher values). This is the relationship the cross-correlation is actually measuring.

The reason we compare burst energy to drawdown rather than price directly is that burst energy measures stress, and drawdown is the natural stress metric for a price series. A market at all-time highs has zero drawdown regardless of its price level. The comparison reveals whether topological stress is aligning with price stress, and the lead/lag offset tells you which one is moving first.

Section 3: The Lead/Lag Structure

This section reveals the temporal relationship between topology and price. Is the topology warning you about something that has not happened yet, or recording something that already has?

Rolling Lead/Lag Timeline



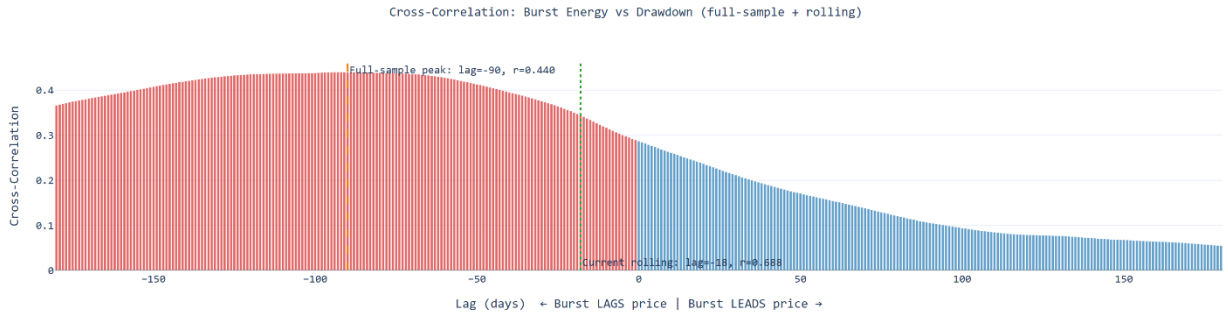
This chart has two panels.

Top panel: A filled area chart showing the smoothed lead/lag value over time. **Blue fill (above zero)** means the topology is leading price by that many days. **Gold fill (below zero)** means it is lagging. The S&P 500 is overlaid as a dotted red line on the right axis for context. Look at how the lead/lag regime relates to major market events — you will often see the topology shift from leading to lagging around major inflection points.

Bottom panel: The cross-correlation at the peak lag, smoothed over 42 trading days. This tells you how strong the burst-price relationship is over time. When this value is near +1 or −1, the topology is tightly coupled to price. When it drifts toward zero, the relationship is weak. The zero line is the critical reference — the correlation frequently crosses zero, and these crossings often coincide with regime transitions.

> **The alternation between leading and lagging**
>
> The market moves through a cycle: the topology tightens (burst energy rises), then price follows (the topology was leading). After a major shock, the topology stays active while price stabilizes (the topology is lagging, registering persistent structural damage). The alternation between these regimes encodes the market's position in the shock-recovery cycle. A persistent leading regime suggests the topology is ahead of the curve. A persistent lagging regime suggests the market is still processing a past event.

Full-Sample Cross-Correlation



This bar chart shows the cross-correlation function computed over the entire dataset. Each bar represents one lag offset, from −180 days (topology lags price) to +180 days (topology leads price). Red bars are negative lags, blue bars are positive.

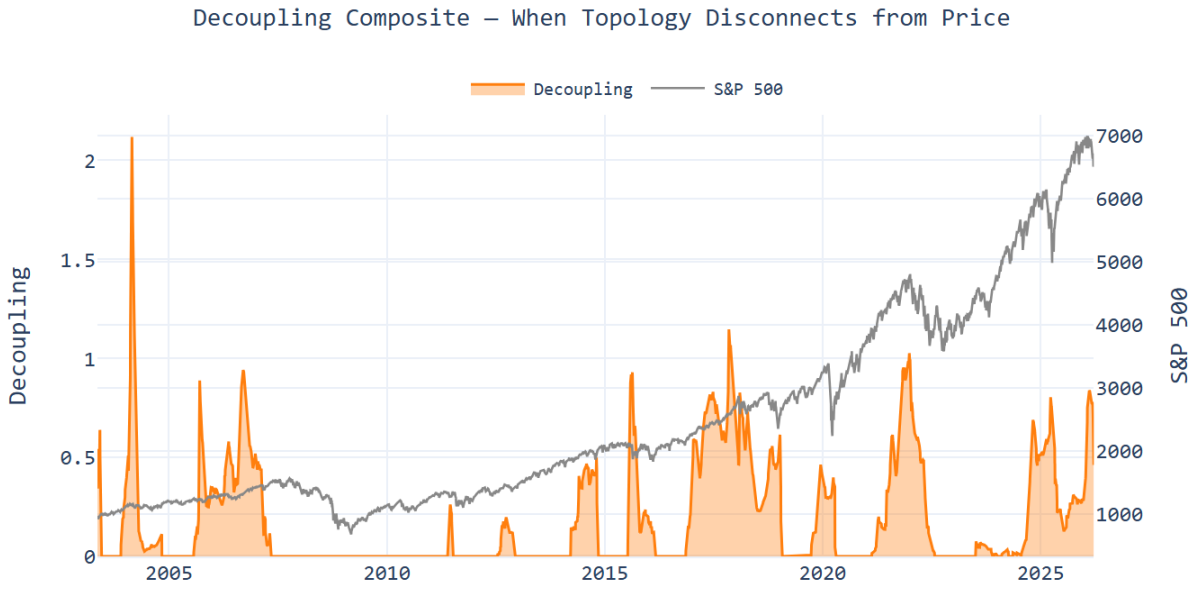
The orange dashed line marks the peak — the lag offset with the strongest correlation. This is the "best fit" alignment of the full dataset. If the peak is at, say, +40 days with a correlation of 0.7, it means that across the entire history, burst energy changes 40 days before corresponding S&P 500 drawdown changes, with a strength of 0.7.

This chart gives you the big-picture structural relationship. The rolling timeline above shows how that relationship evolves; this chart shows the aggregate.

Section 4: The Decoupling Gradient

This section focuses on the most actionable signal in the system: decoupling. When the topology disconnects from price, it historically signals the worst outcomes.

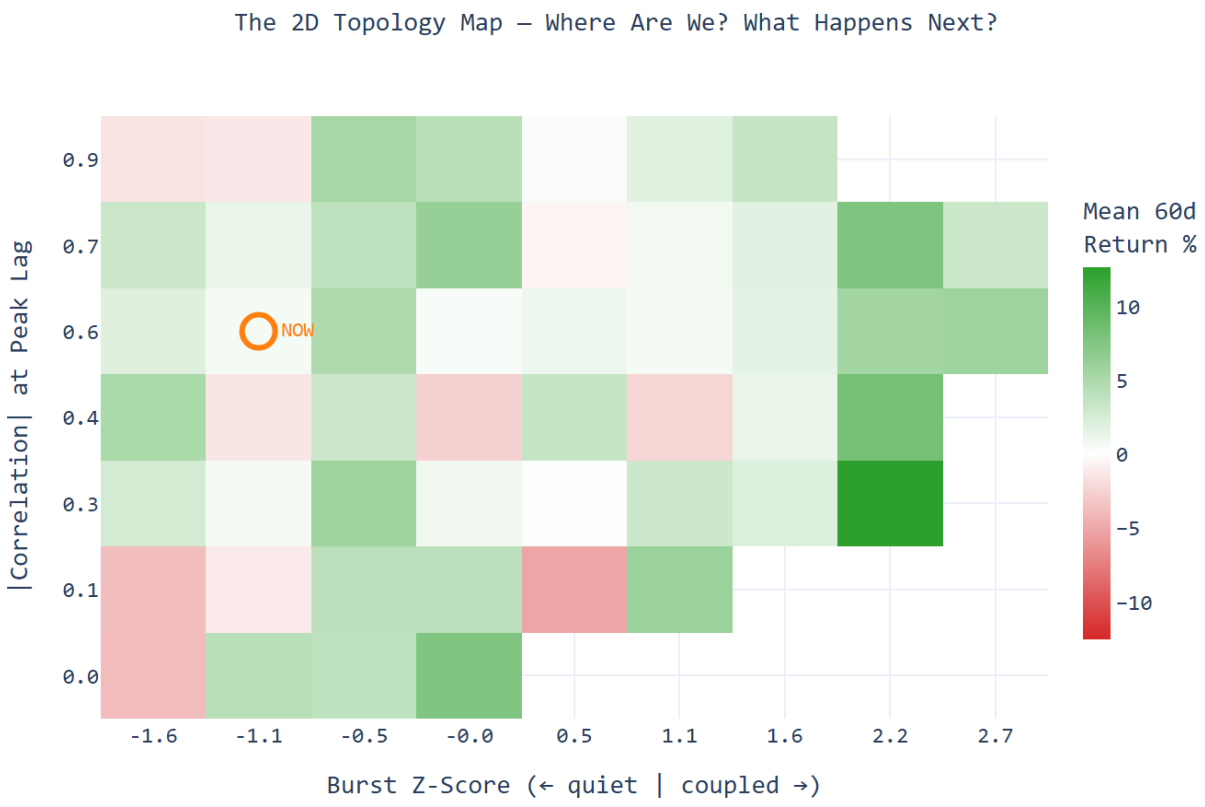
Decoupling Composite Timeline



This chart plots the Decoupling Composite (orange, left axis) against the S&P 500 (gray, right axis) over the full dataset. The decoupling composite is the product of two things: how quiet the topology is (low burst z-score) and how weak the price linkage is (low correlation).

The orange filled areas show when decoupling was elevated. Notice how the major spikes tend to cluster around or just before significant market declines. The composite was elevated before the 2008 crisis, during the 2015–2016 turbulence, and during the 2018 and 2020 corrections. When this signal is flat at zero, it means either burst energy is above average or the correlation is strong — both reassuring.

The 2D Topology Map



This is the most information-dense chart on the dashboard. It is a heatmap with burst z-score on the horizontal axis and |correlation| on the vertical axis. Each cell is colored by the mean forward return of the S&P 500 over the middle cone horizon (60 trading days by default) from that topological state.

Green cells are states that historically produced positive forward returns. **Red cells** are states that produced negative forward returns. White cells are neutral.

The orange circle with "NOW" marks the current position on the map. This tells you at a glance: where are we in the topological state space, and what does history say about outcomes from here?

Reading the axes: the horizontal axis runs from low burst z-score (quiet topology) on the left to high burst z-score (active topology) on the right. The vertical axis runs from low |correlation| (weak price linkage) at the bottom to high |correlation| (strong linkage) at the top.

The most dangerous zone is the lower-left corner: low burst z-score AND low correlation. This is where the topology is both quiet and disconnected. The safest zone is typically the upper portion: high correlation (strong price linkage) regardless of burst z level, because a strong linkage means the adjustment has high confidence.

> **Using the map**
>
> The 2D map is a historical fingerprint. It tells you: from this exact combination of topological activity and price coupling, what has the market done over the next 60 trading days, on average? It does not predict the future — it maps the current state to a distribution of historical outcomes. If the "NOW" marker is in a deep red zone, it does not guarantee a decline, but it means the statistical weight of history is unfavorable.

Section 5: The Signature Library (added v2.0)

Dashboard v2.0 adds an entirely new analytical layer on top of the original four sections. Everything described above still works exactly the same way. The v2.0 additions take the raw topology, lead/lag, and correlation signals that v1.0 already computes and ask a further question: are these signals combining into recognizable patterns that have historically preceded specific outcomes?

The answer is yes. Five distinct topological signatures have been identified and validated across multiple observation horizons. Each signature is a specific geometric configuration of the lead/lag, correlation, and burst energy signals that tends to appear before or during a particular market regime. The signature engine detects these patterns automatically and reports them on the dashboard.

> **Why this is different from traditional technical analysis**
>
> Traditional indicators like MACD, RSI, or moving average crossovers look at price patterns. They measure what prices have done and try to extrapolate. The topological signatures here measure something entirely different: the geometric structure of how hundreds of stocks are cycling relative to each other. A Floor Loss signature, for instance, does not care what the S&P 500 price is doing. It fires when the internal geometry of the market is deteriorating in a specific way that has historically preceded corrections. The market can be making new highs and a Floor Loss can still be active, because the internal structure is weakening even while the headline number looks fine.

The Five Signatures

Each signature is a named pattern built from the same signals the v1.0 dashboard already tracks: lead/lag, correlation, and burst energy. What is new is that the system now looks for specific combinations of these signals that have historically been meaningful.

S1: Floor Loss (Danger Signal)

The most operationally important signature. Floor Loss fires when three things happen simultaneously: the lag is deepening (topology is falling further behind price), decoupling is rising (the connection between topology and price is weakening), and correlation is deteriorating. Think of it as the structural floor giving way underneath the market. Prices may still look normal, but the internal geometry is coming apart. Historically, periods where Floor Loss is active have preceded the most significant corrections in the dataset.

Unlike a binary on/off indicator, Floor Loss has a continuous trigger score from 0.0 to 1.0 that measures how intensely the pattern is present. The score is the geometric mean of three normalized components: how deep the lag is, how elevated the decoupling is, and how much the correlation has deteriorated. This score maps to a taxonomy of five levels: Neutral (0.0–0.25), Structurally Unsupportive (0.25–0.45), Trigger Watch (0.45–0.65), Correction-Prone (0.65–0.80), and Active Damage (0.80–1.0). The structural state badge at the top of the v2.0 dashboard reflects this taxonomy.

S2: Corr Flip (Base Layer Signal)

Corr Flip detects the moment when the correlation between topology and price crosses from positive to negative. This is a regime transition marker. When correlation is positive, higher burst energy corresponds to larger drawdowns (the topology is measuring stress in the normal way). When it flips negative, that relationship inverts. The flip itself is the signal — it marks a structural change in how the market's internal geometry relates to its price. On the dashboard, S2 appears as a discrete event marker (triangle) on the price chart.

S3: Coupled Cascade (Danger Signal)

Coupled Cascade is the crisis-mode signature. It fires when burst energy is high (the topology is extremely active), the lag is deep (topology is behind price), and correlation is negative. This combination means the market is in active structural damage — the topology is registering intense synchronization, but that synchronization is working against price rather than with it. This signature was present during the worst phases of the 2008 financial crisis and the 2020 COVID crash. Like S1, it persists as long as the conditions hold rather than firing as a discrete event. On the dashboard, it appears as a purple band behind the S&P 500 price line.

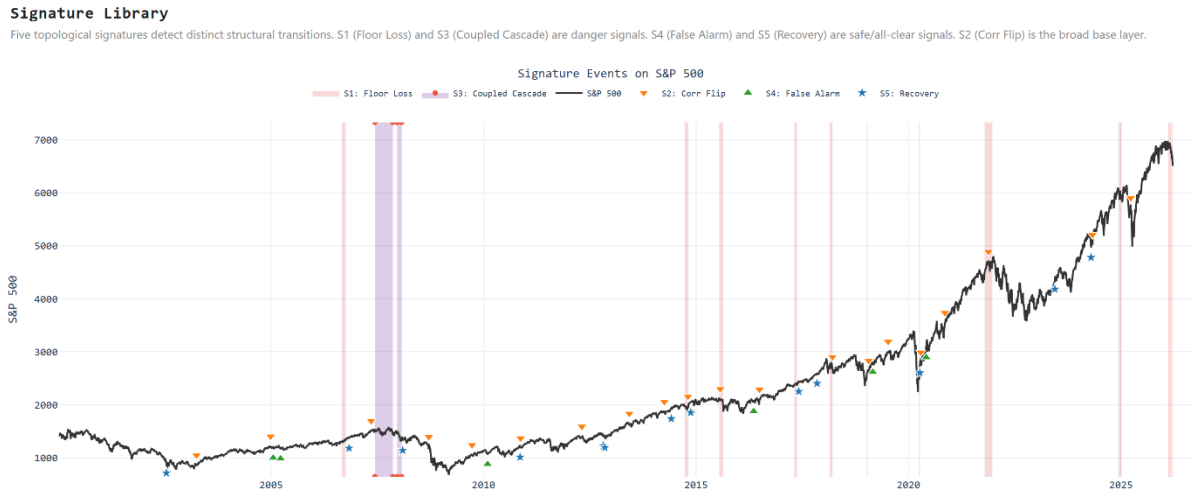
S4: False Alarm (Anti-Signal)

False Alarm is unusual because it dampens other signals rather than raising its own alarm. It fires when the topology flips from leading to lagging while burst energy is elevated (z-score above 0.5). This combination often appears during regime transitions that look threatening but resolve without a major correction. When S4 is active, it reduces the weight given to S1, preventing the system from overreacting to structural shifts that are part of normal market cycling rather than genuine deterioration.

S5: Recovery (All-Clear Signal)

Recovery fires when the topology flips from lagging back to leading while correlation is positive. This means the market's internal structure has normalized — the topology is once again warning ahead of price, and the relationship between topology and price is intact. After a period of Floor Loss or Coupled Cascade, a Recovery signal is the system's way of saying the structural damage has been repaired. On the dashboard, S5 appears as a star marker on the price chart.

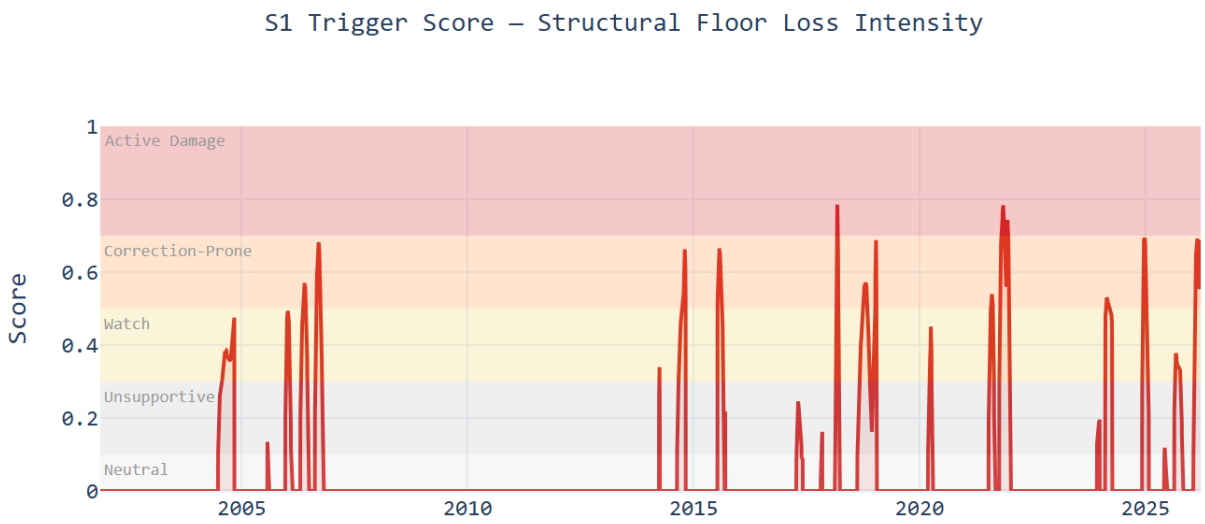
The Signature Timeline Chart



This chart overlays all five signatures onto the S&P 500 price chart. The two state signatures (S1 Floor Loss and S3 Coupled Cascade) appear as colored bands behind the price line: red for Floor Loss, purple for Coupled Cascade. When one of these bands appears, it means that signature was active during that period. The three discrete signatures (S2 Corr Flip, S4 False Alarm, S5 Recovery) appear as triangle or star markers at the dates they fired.

Every element in the legend is clickable. You can toggle individual signatures on and off to isolate specific patterns. For example, clicking to hide everything except S1 shows you exactly when the Floor Loss pattern was active relative to the S&P 500 price history. This is powerful for building intuition about what market environments correspond to each signature.

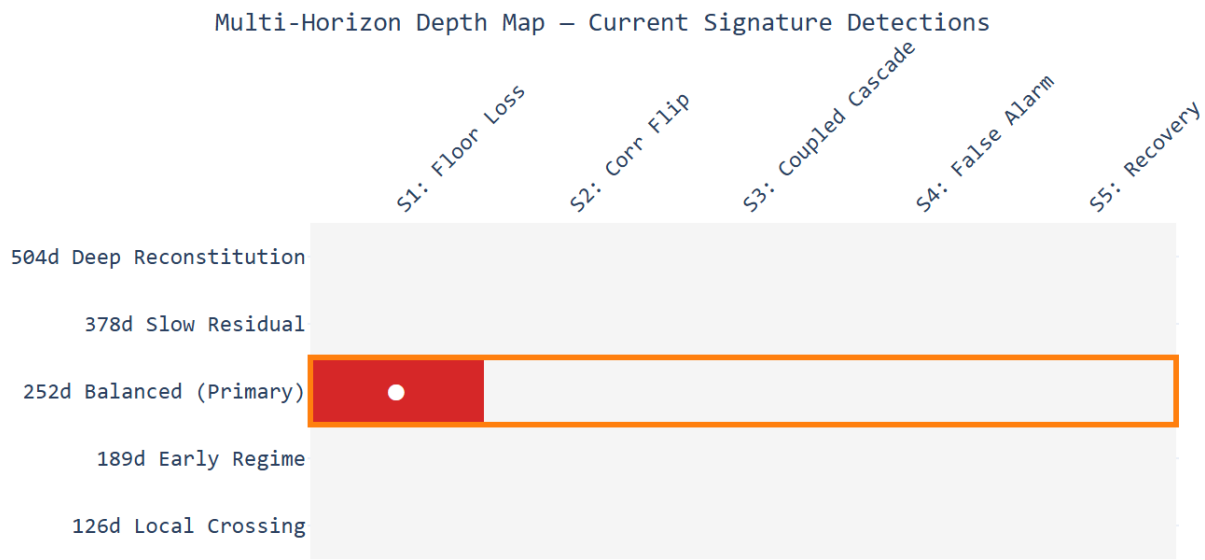
The Trigger Score Chart



This chart plots the S1 Floor Loss trigger score over time. Unlike a simple on/off indicator, the trigger score is a continuous value from 0.0 to 1.0 that measures how strongly the Floor Loss pattern is present. A score of 0.1 means the pattern is barely detectable. A score of 0.85 means the pattern is intense. The chart is colored by the taxonomy bands: green for Neutral, yellow for Structurally Unsupportive, orange for Trigger Watch, red for Correction-Prone, and dark red for Active Damage.

The trigger score is the single most actionable number in the v2.0 dashboard. While the v1.0 dashboard gave you raw signals that required interpretation, the trigger score synthesizes three signals into one number with clear operational meaning. When it enters Correction-Prone territory (above 0.65), the topology is in a configuration that has historically preceded corrections. When it is in Neutral territory (below 0.25), the system is not detecting structural stress.

The Multi-Horizon Depth Map



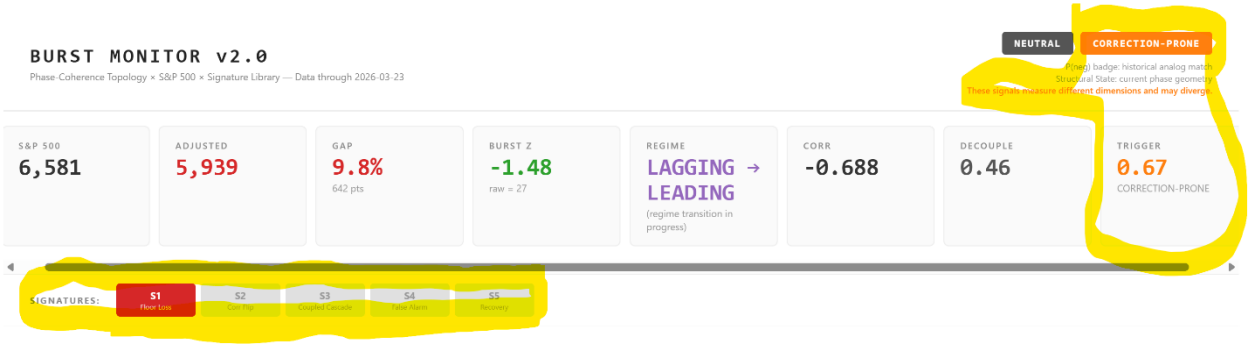
This is perhaps the most conceptually novel chart on the dashboard. The depth map is a grid with five columns (the five signatures, S1 through S5) and five rows (the five closure class horizons: 126 days, 189 days, 252 days, 378 days, and 504 days). Each cell is either filled (signature detected at that horizon) or empty (not detected).

The key insight is that different observation windows resolve different structural layers of the market. The 126-day window sees fast, local crossings. The 252-day window (one trading year) sees the most balanced and interpretable view. The 504-day window (two trading years) sees slow, deep structural processes that persist long after shorter windows have cleared.

When a signature appears at only one horizon, it may be a local, transient pattern. When the same signature appears across multiple horizons simultaneously — what we call concordance — it has structural depth. A Floor Loss that is active at 252 days, 378 days, and 504 days is a much more serious signal than one that is only active at 252 days. The depth map gives you this multi-resolution view at a glance.

> **Reading the depth map**
>
> An empty grid means no signatures are firing at any horizon — the topology is in a clean, neutral state. A single filled cell (say, S1 at 252d) means the Floor Loss pattern is present at the baseline resolution but not confirmed at other horizons. A column of filled cells (S1 active at 252d, 378d, and 504d) means the pattern has multi-horizon concordance — it is present across multiple observation timescales and is structurally deep. The 252d row is the primary reading; other rows confirm or moderate it.

The Enhanced Header and Structural State Badge



In v2.0, the status badge at the top of the dashboard changes from the P(60d neg)-based assessment to the structural state derived from the signature engine. The five levels are: Neutral (no signatures of concern active), Structurally Unsupportive (weak pattern present but not operationally significant), Trigger Watch (pattern present and intensifying), Correction-Prone (strong Floor Loss pattern, historical analogs show elevated correction risk), and Active Damage (the most intense reading, typically seen during or immediately before significant drawdowns).

Below the badge, five small indicator lights labeled S1 through S5 show which individual signatures are currently active at the baseline (252-day) horizon. A lit indicator means that signature's conditions are currently met. The TRIGGER stat card replaces the P(60d NEG) card from v1.0 and shows the current S1 trigger score along with its taxonomy band.

The Five Closure Classes

The signature engine does not run at just one timescale. It evaluates every signature at five different observation windows, called closure classes. These come from the Resolution-Dependent Lead/Lag Addendum, which discovered that different window lengths resolve different structural layers of the market:

126 days (Local Crossing): Fast, partial closures. Frequent sign flips. Useful as a noise filter — if a signature fires here but nowhere else, it is likely transient.

189 days (Early Regime): Crisis intervals become recognizable at this scale. An intermediate resolution.

252 days (Balanced / Primary): One trading year. This is the baseline horizon where all five signatures reach peak clarity. It is the primary reading for operational use.

378 days (Slow Residual): Most intervals at this scale show lag. Used as a confirmation layer — if S1 is active here as well as at 252d, the pattern has depth.

504 days (Deep Reconstitution): Two trading years. Nearly everything at this scale is lag. This window measures the slow structural winding that remains long after shorter-horizon signals have cleared. A signature active here indicates deep, persistent structural change.

Reading the Dashboard as a Whole

The dashboard is designed to be read top-down, from the practical headline to the structural detail. Here is the recommended reading order:

1. Check both status badges. The Structural State badge (v2.0) is your primary one-second assessment — if it reads Neutral, the phase geometry is clean. If it reads Trigger Watch, Correction-Prone, or Active Damage, read further. Then glance at the P(neg) badge to see whether the historical analog record agrees or diverges. Check the S1–S5 indicator lights to see which specific signatures are active.
2. Scan the stat cards. Look at the GAP (is the adjusted price above or below actual?), BURST Z (is the topology active or quiet?), REGIME (leading or lagging — and is there a transition arrow?), DECOUPLE (is the composite elevated?), and the TRIGGER score (what is the S1 Floor Loss intensity?). If P(60d NEG) shows a confidence qualifier, note whether it is in the tail regime or the low-confidence range.
3. Look at the headline chart. Is the adjusted line diverging from actual? Is the cone widening? Are the adjustment percentages (bottom panel) trending positive or negative?
4. Check the lead/lag. Is the topology leading or lagging? A lagging regime during a decline means the topology is still processing the shock. A leading regime suggests it may be anticipating the next move.
5. Look at the decoupling section. If the composite is elevated and the 2D map’s "NOW" marker is in a red zone, that is the strongest warning the system can give. If decoupling is zero and the map is green, the topology is not concerned.

What This System Is Not

This is not a trading signal generator. It does not tell you to buy or sell. It is a structural monitoring system that reveals a dimension of market behavior invisible to traditional analysis. Think of it as an X-ray of the market’s internal geometry. An X-ray can show a fracture, but it does not tell you how to treat it.

The probabilities shown are historical frequencies, not predictions. A P(60d NEG) of 45% means that 45% of the time the market was in a similar structural state, it declined over the following quarter. That is meaningful information, but it is not certainty.

The burst-adjusted S&P 500 is a model output that reflects what the topology implies about fair value. It is one input among many that an informed observer might consider. It is not the "true" price of the market.

Glossary

Burst Energy

The L2 norm of the phase histogram deviation during closure events. Measures the intensity of synchronization across S&P 500 constituents.

Burst Z-Score

Burst energy normalized against its trailing one-year (252-day) average. Zero is average. Negative means quieter than usual. Positive means more active.

Closure Event

A trading day when an unusually large number of stocks complete their price cycles simultaneously, as detected by the phase-coherence method.

Confidence Cone

Percentile bands of historical forward returns from similar topological states. The cone widens when analogs are dispersed and narrows when they agree.

Correlation (at Peak Lag)

The Pearson correlation between burst energy and inverted drawdown at the lag offset that maximizes the relationship. Ranges from −1 to +1.

Decoupling Composite

max(0, −burst_z) × (1 − |correlation|). Measures the degree to which the topology has disconnected from price. Zero is normal; higher is worse.

Drawdown

The percentage decline from the all-time high of the S&P 500 at any given point. Used as the stress measure against which burst energy is compared.

Discrimination Threshold

The decoupling composite value (0.3) above which the analog pool becomes meaningfully concentrated and the P(neg) probability reading gains actionable signal strength. Below this threshold, most of the market's history is included in the analog pool, producing readings close to the unconditional base rate.

Hilbert Transform

A mathematical operation that extracts the instantaneous phase (cycle position) of each stock's price series.

Lead/Lag

The time offset (in trading days) at which burst energy best correlates with S&P 500 drawdown. Positive = topology leads price. Negative = topology lags.

L2 Norm

The Euclidean length of a vector. Applied to the phase histogram deviation, it measures how far the observed distribution is from uniform (random).

Phase

A stock's position within its current price cycle, measured from 0 to 1. Extracted using the Hilbert transform on the detrended price series.

Phase-Coherence

The condition where many stocks are at the same point in their respective cycles simultaneously. High coherence = high burst energy.

Regime (Leading / Lagging)

The current state of the topology's temporal relationship with price, determined by the sign of the smoothed lead/lag value.

Regime Transition

A state where the raw (unsmoothed) lead/lag value has changed sign but the 21-day smoothed value has not yet followed. Indicates the topology's temporal relationship with price is in the process of flipping from leading to lagging or vice versa. The dashboard shows an arrow notation and a blended color during this period, which typically lasts one to three weeks.

Topology

In this context, the hidden geometric structure formed by the relationships between stocks' phase cycles. Not directly observable in price data.

Closure Class

One of five observation window lengths (126, 189, 252, 378, 504 trading days) at which the signature engine evaluates the lead/lag and correlation structure. Each closure class resolves a different structural layer of the market.

Concordance

When the same signature is detected simultaneously across multiple closure class horizons. Multi-horizon concordance indicates a signal with structural depth rather than a transient, local pattern.

Cone Confidence

A qualifier (HIGH or LOW) indicating whether the confidence cone and P(neg) readings are based on a differentiated analog pool. HIGH (cone_confidence) when the decoupling composite is at or above the discrimination threshold of 0.3. LOW when below, meaning the analog pool is undifferentiated and the probability reading is close to the unconditional base rate.

Structural State

The composite assessment produced by the v2.0 signature engine. Five levels: Neutral, Structurally Unsupportive, Trigger Watch, Correction-Prone, Active Damage. Derived from the S1 trigger score and multi-horizon concordance.

Trigger Score

A continuous value from 0.0 to 1.0 measuring the intensity of the S1 Floor Loss signature. Computed as the geometric mean of three normalized components: lag depth, decoupling intensity, and correlation deterioration.

Signature (Topological)

A named pattern of lead/lag, correlation, and burst energy signals that has been validated as historically meaningful. Five signatures are defined: S1 Floor Loss (danger), S2 Corr Flip (regime transition), S3 Coupled Cascade (crisis), S4 False Alarm (anti-signal), S5 Recovery (all-clear).

Appendix A: The Logic of Phase-Coherence Topology

This appendix explains the reasoning behind the Burst Monitor's approach. It is written for a reader who has worked through the dashboard sections above and wants to understand why the system is built this way — what problem it solves, what assumptions it rests on, and why the specific mathematical choices were made. It is not a substitute for the published research papers, but it covers the essential logic at the level needed to interpret the signals with confidence.

Why Phase, Not Price

Most quantitative approaches to understanding market structure start from price returns. Correlation matrices, principal component analysis, beta coefficients, volatility clustering — all of these take price movement as their raw input. The Burst Monitor does not. It starts from phase, and this choice is not a stylistic preference. It addresses a specific failure mode in return-based methods that matters most exactly when you need those methods to work.

The failure mode is what the research calls the amplitude blind spot. Consider two situations. In the first, a major news event hits and every stock moves sharply in the same direction. Correlation spikes. In the second, stocks have been gradually synchronizing their business cycles — they are peaking and troughing together, completing their oscillations in lockstep — but without dramatic price moves on any single day. Return-based correlation captures the first situation easily but is nearly blind to the second. Yet the second situation is arguably more informative, because it reveals that the market's degrees of freedom have collapsed. The stocks are no longer doing independent things. That structural loss of independence is what precedes and enables the cascade, not the cascade itself.

To see this structural coupling without contamination from price amplitude, the system uses the Hilbert transform. This is a standard signal processing operation that extracts the instantaneous phase of a time series — where the signal sits within its current oscillation cycle. Applied to a detrended stock price, it returns an angle that moves from 0 to 1 as the stock completes a business cycle. Critically, a stock that moves 1% in a day and a stock that moves 10% in a day can produce identical phase readings if they are at the same point in their respective cycles. Amplitude is stripped out entirely. What remains is pure cyclical position.

This is the foundational commitment of the system: market structure is better read in the phase domain than the amplitude domain, because phase reveals the geometry of coupling while amplitude reveals only the energy of movement.

The Unit Circle and Closure Events

Once every stock has a phase coordinate between 0 and 1, the entire market can be visualized as a collection of points on a circle (or, more precisely, a torus, since each stock's phase is periodic). On any given trading day, these points are distributed around the circle according to where each stock sits in its cycle.

If the market were truly high-dimensional — if every stock were doing its own independent thing — these points would be scattered approximately uniformly around the circle. No clustering, no structure. The phase histogram would be flat.

A closure event is the opposite of this. It occurs when an unusually large number of stocks arrive at the same phase simultaneously — specifically, near the origin (phase close to 0 or 1, meaning they are completing a cycle). The system detects this by counting how many stocks have phase values within 0.05 units of the origin. If that count exceeds a threshold — at least 5 stocks, or 8% of the current universe, whichever is larger — the day is flagged as a closure event.

The threshold scales with universe size for a geometric reason. The original research used a fixed basket of 59 stocks, where 5 closing simultaneously was structurally meaningful. When the system expanded to rolling S&P 500 membership (243 to 504 constituents), the threshold had to scale proportionally to preserve the same geometric meaning: a fixed fraction of the circle's occupants converging on one point.

On closure days, the system measures how deformed the rest of the circle is. It takes the phase histogram of the non-closing stocks (using 36 bins around the circle), compares it to the long-run background histogram across all dates, and computes the L2 norm — the Euclidean distance between the two histograms. A high L2 norm means the non-closing stocks are also bunched up somewhere, not uniformly distributed. The market's phase space has collapsed from a high-dimensional configuration into a low-dimensional one. This L2 norm is the raw burst energy.

The choice of L2 norm (rather than, say, KL divergence or total variation distance) is deliberate. L2 is sensitive to the shape of the deviation, not just its magnitude. Two histograms that deviate from uniform in geometrically different ways produce different L2 values even if their total deviation is the same. This makes burst energy a measure of geometric deformation, not just statistical departure from randomness.

From Closure Events to a Structural Intensity Series

Raw closure events are sparse and spiky. On most trading days, the closure threshold is not met, so the raw burst energy is zero. On closure days, it can spike to very different levels. This makes the raw series difficult to interpret as a continuous signal.

The smoothing procedure converts these sparse events into a continuous structural intensity reading. For every trading day (not just closure days), the system looks back over a rolling 252-day window (one trading year) and computes the mean of the top-decile burst energy values in that window. This is a critical design choice for two reasons.

First, using the top decile rather than a simple average ensures that the smoothed value reflects the strongest closure events in the lookback period, not the average strength of all events. Structural coupling is an extreme phenomenon — it is the tail of the distribution that carries information, not the center.

Second, the computation runs on every trading day, not just closure days. Earlier versions of the pipeline used forward-fill from the last closure day, which created staircase artifacts — the smoothed value would hold flat for days and then jump. Computing the rolling window every day ensures a smooth, continuously updating intensity series that properly decays as old closure events roll out of the window.

The resulting burst energy series is then standardized as a z-score against its own expanding historical mean and standard deviation. This is the Burst Z-Score on the dashboard: how unusual the current structural coupling is relative to everything the system has observed so far.

The IAAFT Null Hypothesis

Any claimed signal must be tested against a null hypothesis. The question here is: could the burst energy pattern we observe be produced by stocks that have the same individual statistical properties but no cross-asset phase alignment?

The system answers this with Iterative Amplitude-Adjusted Fourier Transform (IAAFT) surrogates. This is a method from nonlinear dynamics (Schreiber and Schmitz, 1996) that generates synthetic time series which preserve both the exact power spectrum and the amplitude distribution of each individual stock. What the surrogates destroy is the phase relationships between stocks. Each surrogate stock has the same autocorrelation structure and the same distribution of returns as the real stock, but its phase is scrambled relative to every other stock.

The system generates 100 such surrogate ensembles, runs the full burst energy pipeline on each one, and constructs a 99% confidence envelope. This envelope represents the range of burst energy you would expect to see if the cross-asset phase structure were random — if the apparent closure events were just coincidence arising from each stock's individual dynamics.

The key validation result: across 2,966 trading day observations, the real market's burst energy never once intersected the 99% surrogate envelope. The mean signal-to-null ratio was 2.59 times. This means the phase structure of the real market is categorically different from what you would get by chance, even after accounting for each stock's own cyclical behavior. The closure events are not artifacts of individual stock dynamics. They are a genuine cross-asset phenomenon.

Lead, Lag, and the Resolution Boundary

The burst energy series captures the intensity of phase-coherent structural coupling. The next question is temporal: does this structural coupling anticipate price movement, or follow it?

The answer, documented in the Lead/Lag Addendum, is both — in alternating regimes. The system computes a rolling cross-correlation between burst energy and the inverted S&P 500 drawdown in 252-day windows. At each window, it extracts the lag that maximizes the absolute correlation. When the best lag is positive, topology is leading price: structural coupling is tightening before the drawdown materializes. When the best lag is negative, topology is lagging: prices have bounced but the phase structure remains compressed.

This alternation has a structural interpretation. In the leading regime (blue on the dashboard), phase coupling is tightening before the drawdown arrives. The market's internal diversification is degrading, but prices have not yet reflected this. This is the approach phase of a crisis. In the lagging regime (gold), prices have recovered — often via central bank intervention or liquidity injection — but the phase structure remains compressed. Business cycles have not desynchronized yet. The recovery is structurally incomplete.

A full-sample cross-correlation misleadingly suggests that topology always lags (peak at -30 days, r = 0.261). But the broad, flat plateau shape of the cross-correlation function reveals a signal that alternates between regimes. The rolling decomposition confirms this: roughly half the time the topology leads, and half the time it lags, at the baseline 252-day window.

The most conceptually important finding is the resolution boundary. The system evaluates lead/lag at five different observation windows: 126, 189, 252, 378, and 504 trading days. These are the five closure classes. At shorter windows (126 days), the lead/lag relationship flips frequently — the system sees fast, local crossings. At 252 days, the alternation between leading and lagging is at its most interpretable. At longer windows (378, 504 days), nearly everything appears as lag.

The transition from balanced alternation to lag dominance occurs between 280 and 315 trading days. This is not an arbitrary boundary — it corresponds to the timescale of the system's own recovery dynamics. The macro-shock decay times (from the GFC through the 2022 rate hikes) cluster in the 152-to-364 day range. When the observation window crosses this range, it is looking at intervals long enough to contain an entire shock-and-recovery cycle, which makes nearly everything appear as lag. The resolution boundary is the system observing its own recovery timescale.

The Signatures as Geometric Events

The five signatures (S1 through S5) are not pattern-matching rules imposed on the data from outside. They are geometric conditions in the phase-coupling space defined by three axes: lead/lag direction and depth, correlation strength and sign, and burst energy level. Each signature names a specific configuration in this space that has been validated as historically meaningful across all five closure classes.

S1 (Floor Loss) is the most operationally important. It fires when three things happen simultaneously: the lag is deepening (the topology is falling further behind price), the decoupling composite is rising (the topology is going quiet while disconnecting from price), and the correlation is deteriorating (the remaining coupling is weakening). This is a compound geometric condition — not just one axis moving, but three axes converging on a specific corner of the state space. The continuous trigger score (0.0 to 1.0) measures how far into that corner the system currently sits, computed as the geometric mean of three normalized components. This is why it is called a trigger score rather than a binary signal: the geometry of approach matters, not just whether a threshold was crossed.

S2 (Corr Flip) marks the moment when correlation crosses from positive to negative. In isolation, this is a base-layer regime transition marker. Its importance lies in what it means geometrically: the relationship between phase coupling and price stress has reversed sign. The topology and price are now moving in opposition rather than in concert.

S3 (Coupled Cascade) fires during active crises: high burst energy (the topology is extremely active), deep lag (the topology is behind price), and negative correlation. This is the geometric signature of a macro shock in progress — the phase structure is both intense and damaged. S1 detects the approach; S3 detects the event itself.

S4 (False Alarm) is an anti-signal. It fires when the topology flips from leading to lagging while burst energy is elevated ($Z > 0.5$). This combination — a regime flip during active coupling — turns out to be geometrically benign: the topology is rearranging itself, not deteriorating. S4's presence dampens S1, preventing false positives during these structural transitions.

S5 (Recovery) fires when the topology flips from lagging back to leading while correlation is positive. This is the geometric all-clear: the phase structure has normalized, business cycles have desynchronized, and the topology is once again anticipating rather than trailing price.

The multi-horizon depth map reveals whether a signature has structural depth. When S1 appears at the 252-day window alone, it may be a transient local pattern. When S1 appears simultaneously at 252 and 378 days, it means the deterioration is visible across multiple structural timescales — it has geometric depth, not just surface activity. This concordance across closure classes is the primary mechanism for escalating the structural state from Neutral through Structurally Unsupportive, Trigger Watch, Correction-Prone, and Active Damage.

Hyper-Elasticity and the Decay Sequence

The final piece of the theoretical picture is an observation about how the market's structural response to shocks has changed over time. When a macro shock hits — a financial crisis, a geopolitical event, a pandemic — burst energy spikes as phase coupling intensifies. The question is: how long does the topology take to recover? How many trading days until the phase structure returns to a high-dimensional, desynchronized state?

The decay times, measured from the burst energy peak to recovery baseline, form a striking sequence. The 2008 Global Financial Crisis took 364 trading days. The 2011 Euro Crisis took 291 days. The 2016 Brexit shock took 152 days. COVID in 2020 took 252 days (an amplitude exception — the price move was enormous but structural recovery was faster than the GFC). The 2022 Rate Hike cycle took 173 days.

Setting aside COVID (whose amplitude was historically anomalous), the sequence is near-monotonically decreasing. A permutation test on this ordering yields $p = 0.0085$ — it is statistically unlikely to be random. The market appears to recover structural independence faster with each successive shock. The research calls this the hyper-elastic market hypothesis: the market's connective tissue is becoming more elastic, snapping back to a high-dimensional state more quickly than it used to.

An exponential fit to the decay sequence projects approximately 135 trading days for the next typical-amplitude macro shock. This is a structural prediction, not a price prediction — it says nothing about how far prices will fall, only about how long the phase geometry will remain compressed.

The theory comes with explicit falsification thresholds. If the next macro shock's peak burst energy fails to separate from the 99% IAAFT surrogate envelope, the phase-coherence framework itself is in question. If a typical-amplitude shock takes more than 350 trading days to recover — reverting to pre-2009 structural behavior — the hyper-elasticity hypothesis is falsified. These are not soft caveats. They are specific, testable conditions under which the framework's practitioners should question its continued validity.

This willingness to specify falsification conditions is, in the end, what distinguishes the Burst Monitor from the universe of market indicators that never say when they would be wrong. The system rests on a geometric claim about market structure. That claim is either true or it is not, and the IAAFT validation and decay sequence provide the empirical footing to tell the difference.